

Lecture 24- Static NAT (Network Address Translation)

1 The Problem with IPv4 Addresses

IPv4 uses **32-bit addressing**, which allows about:

 **4.3 billion IP addresses**

At first this seemed enough, but today the internet has billions of devices.

Devices connected to the Internet

-  Smartphones
-  Laptops
-  Servers
-  IoT devices
-  Smart TVs
-  Gaming consoles

Because of this huge number of devices:

 **IPv4 addresses are not enough anymore**

This problem is called **IPv4 Address Exhaustion**.

2 Solutions for IPv4 Address Shortage

There are **long-term** and **short-term** solutions.

 **Long-Term Solution**

 **IPv6**

IPv6 uses **128-bit addressing**, providing an extremely large number of IP addresses.

Example IPv6 address:

2001:0db8:85a3:0000:0000:8a2e:0370:7334

But switching the whole internet to IPv6 takes time.

Short-Term Solutions (Important for CCNA)

Before IPv6 adoption, three techniques helped extend IPv4 usage:

- 1 CIDR (Classless Inter-Domain Routing)**
- 2 Private IPv4 Addresses**
- 3 NAT (Network Address Translation)**

 In this lecture we focus on **NAT**.

3 Private IPv4 Addresses (RFC 1918)

Private IP addresses are defined in **RFC 1918**.

These addresses:

-  Can be used inside private networks
-  Cannot be used directly on the Internet

They are commonly used in:

-  Home networks
 -  Company networks
 -  School networks
 -  Data centers
-



Private IP Address Ranges

Class	Network	Range
Class A	10.0.0.0/8	10.0.0.0 – 10.255.255.255
Class B	172.16.0.0/12	172.16.0.0 – 172.31.255.255
Class C	192.168.0.0/16	192.168.0.0 – 192.168.255.255



Interview Tip

A very common interview question:



“What are the private IPv4 ranges?”

You should immediately answer:

- 10.0.0.0 /8
- 172.16.0.0 – 172.31.255.255
- 192.168.0.0 /16



What is NAT (Network Address Translation)?



Definition

NAT (Network Address Translation) is a networking technique where a router translates private IP addresses into public IP addresses.

This allows **devices in a private network to communicate with the internet.**



NAT Basic Idea

Private IP → Router → Public IP → Internet

Example:

172.16.0.1 → Router → 100.0.0.1 → Internet

The router **replaces the private IP with a public IP.**



Daily Life Example of NAT

Imagine a home network.

Devices inside the house

Device	Private IP
--------	------------

Phone	192.168.1.2
-------	-------------

Laptop	192.168.1.3
--------	-------------

Smart TV	192.168.1.4
----------	-------------

But the **Internet Service Provider (ISP)** gives only:



1 public IP address

Example:

49.34.78.120

The home router uses **NAT** so that:

- ✓ All devices share one public IP
- ✓ Internet still works for everyone

So NAT acts like a **translator between private and public networks**.



Why NAT is Used

Reason	Explanation
--------	-------------



Save IPv4 addresses	Many devices use one public IP
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Security	Internal IP addresses are hidden
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Enterprise networks	Large internal networks
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Internet access	Private devices reach public websites
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Types of NAT

There are **three main types of NAT**.

NAT Type	Description
Static NAT	One private IP $\leftrightarrow$ One public IP
Dynamic NAT	Pool of public IPs
PAT (NAT Overload)	Many devices share one public IP

This lecture focuses on **Static NAT**.



Static NAT Explained

Definition

Static NAT creates a **permanent one-to-one mapping** between a private IP and a public IP.

Private IP $\leftrightarrow$ Public IP

Example:

172.16.0.1 $\leftrightarrow$ 100.0.0.1

This mapping **never changes**.



Example Scenario

Suppose a company has a **public web server** inside its network.

The server private IP:

172.16.0.10

But the internet must access it using a **public IP**.

So NAT mapping is created:

172.16.0.10 $\leftrightarrow$ 203.0.113.10

Now users on the internet can access the server.

⚠ Static NAT Disadvantages

- ✗ Wastes public IP addresses
- ✗ Not scalable
- ✗ Not practical for large networks

But it is useful for:

- ✓ Servers
- ✓ Permanent mappings
- ✓ Learning NAT concepts

📖 ? NAT Terminology

🏠 Inside Local

The **private IP address of the internal device.**

Example:

172.16.0.1

This address exists **inside the local network.**

🌐 Inside Global

The **public IP address representing the internal device.**

Example:

100.0.0.1

This address is visible on the **internet.**

Outside Global

The **real public IP of the destination device**.

Example:

8.8.8.8

This is the **actual internet server IP**.

Outside Local

The **destination IP as seen from the internal network**.

Often it is the same as outside global.

NAT Flow (Easy Explanation)

Example communication:

PC → Router → Internet Server

Actual IP translation:

Inside Local: 172.16.0.1

Inside Global: 100.0.0.1

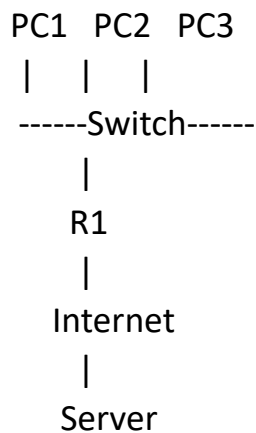
Destination: 8.8.8.8

Flow:

172.16.0.1 → Router → 100.0.0.1 → 8.8.8.8

Static NAT Lab Configuration

Network Topology



Network Details

PCs Network

172.16.0.0/24

Device IP

PC1 172.16.0.1

PC2 172.16.0.2

PC3 172.16.0.3

Gateway:

172.16.0.254

Router Interfaces

Interface	IP
-----------	----

R1 G0/1 (LAN)	172.16.0.254
---------------	--------------

R1 G0/0 (WAN)	203.0.113.1
---------------	-------------

Internet Router	203.0.113.2
-----------------	-------------



1 Configure IP Addresses on PCs

PC1

IP: 172.16.0.1

Mask: 255.255.255.0

Gateway: 172.16.0.254

PC2

IP: 172.16.0.2

Mask: 255.255.255.0

Gateway: 172.16.0.254

PC3

IP: 172.16.0.3

Mask: 255.255.255.0

Gateway: 172.16.0.254



2 Configure Router Interfaces

R1> enable

R1# configure terminal

interface g0/1

ip address 172.16.0.254 255.255.255.0

no shutdown

interface g0/0

ip address 203.0.113.1 255.255.255.252

no shutdown

exit

3 Define NAT Inside and Outside

LAN interface = **Inside**

```
interface g0/1  
ip nat inside
```

Internet interface = **Outside**

```
interface g0/0  
ip nat outside
```

4 Configure Static NAT Mappings

```
ip nat inside source static 172.16.0.1 100.0.0.1  
ip nat inside source static 172.16.0.2 100.0.0.2  
ip nat inside source static 172.16.0.3 100.0.0.3
```

Meaning:

Private IP	Public IP
------------	-----------

172.16.0.1	100.0.0.1
------------	-----------

172.16.0.2	100.0.0.2
------------	-----------

172.16.0.3	100.0.0.3
------------	-----------

5 Save Configuration

do write

or

write memory

6 Verify NAT Table (Before Traffic)

show ip nat translations

Example output:

Inside global	Inside local
100.0.0.1	172.16.0.1
100.0.0.2	172.16.0.2
100.0.0.3	172.16.0.3

At this stage:

✓ Static entries appear

✗ No traffic entries yet

7 Generate Traffic

From **PC1**, ping an internet server:

ping 8.8.8.8

8 Verify NAT Table After Ping

show ip nat translations

Example output:

icmp 100.0.0.1:1 172.16.0.1:1 8.8.8.8:1 8.8.8.8:1

NAT Table Explanation

Field	Meaning
Inside Local	Private IP of device
Inside Global	Translated public IP
Outside Local	Destination seen internally
Outside Global	Real destination IP

Key Observations

- ✓ Static NAT entries exist even without traffic
 - ✓ ICMP entries appear when traffic starts
 - ✓ NAT translation happens at the router
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Advantages and Disadvantages

Advantages

- ✓ Simple configuration
 - ✓ Predictable mapping
 - ✓ Good for servers
-

Disadvantages

-  Consumes many public IP addresses
 -  Not scalable
 -  Not suitable for large networks
-

CCNA Exam Summary

Topic	Key Point
NAT	Translates private IP to public IP
RFC 1918	Defines private IP ranges
Static NAT	One-to-one mapping
Inside Local	Private IP
Inside Global	Public IP
Command	ip nat inside source static
Verification	show ip nat translations



CHAPTER SUMMARY



IPv4 Address Problem

IPv4 uses **32-bit addressing**.

Total IPv4 addresses:

$$2^{32} = 4,294,967,296 \text{ (~4.3 Billion)}$$

Initially this seemed enough, but today billions of devices are connected to the internet.

Devices using IP addresses

-  Smartphones
-  Laptops
-  Servers
-  IoT devices
-  Smart TVs
-  Gaming consoles

Because of this massive growth:

✗ IPv4 addresses started running out

This problem is called:

IPv4 Address Exhaustion



Solutions to IPv4 Exhaustion

There are **two types of solutions**.



Long-Term Solution

IPv6

IPv6 uses **128-bit addressing**.

Example IPv6:

2001:0db8:85a3:0000:0000:8a2e:0370:7334

IPv6 provides **almost unlimited IP addresses**.

However, migrating the entire internet to IPv6 takes time.

Short-Term Solutions

Before IPv6 became widespread, three technologies helped extend IPv4 usage.

1 CIDR (Classless Inter-Domain Routing)

2 Private IP Addressing

3 NAT (Network Address Translation)

This chapter focuses mainly on **NAT**.

Private IPv4 Addresses (RFC 1918)

Private IP addresses are defined in **RFC 1918**.

These addresses:

- ✓ Used inside private networks
- ✗ Cannot be routed directly on the internet

They are commonly used in:

-  Home networks
 -  Corporate networks
 -  Schools
 -  Data centers
-

Private IP Ranges

Class	Network	Range
Class A	10.0.0.0/8	10.0.0.0 – 10.255.255.255
Class B	172.16.0.0/12	172.16.0.0 – 172.31.255.255

Class	Network	Range
-------	---------	-------

Class C	192.168.0.0/16	192.168.0.0 – 192.168.255.255
---------	----------------	-------------------------------

These ranges are **not routable on the public internet**.

What is NAT?

Definition

NAT (Network Address Translation) is a process where a router translates **private IP addresses into public IP addresses**.

This allows private devices to communicate with the internet.

NAT Basic Concept

Private IP → Router → Public IP → Internet

Example

172.16.0.1 → Router → 100.0.0.1 → Internet

The router **replaces the private IP with a public IP**.

Real-Life Example of NAT

Imagine a home network.

Device	Private IP
--------	------------

Phone	192.168.1.2
-------	-------------

Laptop	192.168.1.3
--------	-------------

Smart TV	192.168.1.4
----------	-------------

But the ISP provides only:

 **1 public IP address**

Example:

49.34.78.120

The router performs NAT so:

- ✓ All devices share one public IP
 - ✓ Internet works for everyone
-

Why NAT is Used

Reason	Explanation
 IPv4 conservation	Many devices share one public IP
 Security	Internal IP addresses are hidden
 Enterprise networking	Internal networks remain private
 Internet access	Private networks reach public websites

Types of NAT

There are **three major types**.

NAT Type	Description
Static NAT	One private IP ↔ One public IP
Dynamic NAT	Pool of public IPs
PAT (NAT Overload)	Many devices share one public IP

This lecture focuses on **Static NAT**.

Static NAT Explained

Definition

Static NAT creates a **permanent one-to-one mapping** between a private IP and a public IP.

Example:

172.16.0.10 ↔ 203.0.113.10

This mapping **never changes**.

Example Scenario

A company hosts a **web server** inside its internal network.

Private IP:

172.16.0.10

Public mapping:

203.0.113.10

Now internet users access the server using:

http://203.0.113.10

The router translates it to the internal server.

⚠ Static NAT Disadvantages

- ✗ Consumes many public IP addresses
- ✗ Not scalable
- ✗ Not practical for large networks

But it is useful for:

- ✓ Public servers
 - ✓ Permanent mappings
 - ✓ Learning NAT concepts
-

📖 NAT Terminology

Understanding NAT terminology is **very important for interviews**.

🏠 Inside Local

The **private IP address** of the internal device.

Example:

172.16.0.1



Inside Global

The **public IP address representing the internal device.**

Example:

100.0.0.1



Outside Global

The **real public IP of the destination device.**

Example:

8.8.8.8



Outside Local

The **destination address as seen inside the network.**

Usually the same as outside global.



NAT Flow Example

Communication example:

PC → Router → Internet Server

Translation process:

Inside Local : 172.16.0.1

Inside Global : 100.0.0.1

Destination : 8.8.8.8

Packet flow:

172.16.0.1 → Router → 100.0.0.1 → 8.8.8.8

NAT Table Fields

Field	Meaning
Inside Local	Private IP
Inside Global	Public IP
Outside Local	Destination inside view
Outside Global	Actual destination



CHAPTER CONCLUSION

Static NAT is a **fundamental networking concept** used to connect private networks to the internet.

It solves the **IPv4 exhaustion problem** by translating private IP addresses into public IP addresses.

Key takeaways:

- ✓ Private IPs cannot access the internet directly
- ✓ NAT allows private networks to communicate externally
- ✓ Static NAT creates **permanent one-to-one mapping**
- ✓ NAT is performed by routers
- ✓ NAT tables track translation entries

Even though Static NAT is **not scalable**, it is still widely used for:

- Public servers
 - Email servers
 - Web hosting
 - Firewall NAT rules
-

Q & A

1 What is NAT?

NAT (Network Address Translation) is a networking technique used by routers to translate private IP addresses into public IP addresses so that internal devices can communicate with external networks such as the internet.

It helps conserve IPv4 addresses and improves network security by hiding internal IP addresses.

2 Why was NAT created?

NAT was introduced to solve the **IPv4 address exhaustion problem**.

Because billions of devices require IP addresses, NAT allows many devices to share a limited number of public IP addresses.

3 What are private IP address ranges?

Private IP ranges defined by RFC 1918 are:

10.0.0.0/8

172.16.0.0 – 172.31.255.255

192.168.0.0/16

These addresses are used in internal networks and are not routable on the internet.

4 What are the types of NAT?

Three main types:

Type	Description
Static NAT	One-to-one mapping
Dynamic NAT	Uses pool of public IPs

Type	Description
PAT	Many devices share one IP

5 What is Static NAT?

Static NAT creates a **permanent mapping between a private IP and a public IP**.

Example:

172.16.0.10 ↔ 203.0.113.10

6 What is Inside Local?

Inside Local refers to the **private IP address of the internal device inside the network**.

Example:

172.16.0.1

7 What is Inside Global?

Inside Global is the **public IP address representing the internal device on the internet**.

Example:

100.0.0.1

8 What is Outside Global?

Outside Global is the **real public IP address of the external server**.

Example:

8.8.8.8

9 What is Outside Local?

Outside Local is the **destination IP address as seen by the internal network**.

Often it is the same as the outside global address.

10 Which command configures Static NAT?

`ip nat inside source static [private-ip] [public-ip]`

Example:

`ip nat inside source static 172.16.0.1 100.0.0.1`

1 1 How do you verify NAT configuration?

`show ip nat translations`

This command displays NAT translation entries.

1 2 What are NAT inside and outside interfaces?

Inside interface:

`ip nat inside`

Outside interface:

`ip nat outside`

Inside interface connects to **private network**, outside connects to **public network**.

1 3 What happens when NAT translation occurs?

The router replaces the **source private IP with a public IP** before sending packets to the internet.

1 4 What are advantages of NAT?

- ✓ Conserves IPv4 addresses
 - ✓ Hides internal network structure
 - ✓ Improves security
 - ✓ Allows multiple devices to share one public IP
-

1 5 What are disadvantages of Static NAT?

- ✗ Requires many public IP addresses
 - ✗ Not scalable
 - ✗ Complex management in large networks
-